ERGA Assembly Report

v24.04.03_beta

Tags: ERGA-BGE

TxID	3102972
ToLID	**qmTriMatu1**
Species	Trichoniscus matulici
Class	Malacostraca
Order	Isopoda

Genome Traits	Expected	Observed
Haploid size (bp)	2,904,951,799	2,869,951,676
Haploid Number	8 (source: ancestor)	8
Ploidy	2 (source: ancestor)	2
Sample Sex	F	F

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 5.8.Q52

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Assembly length loss > 3% for pri
. More than 1000 gaps/Gbp for pri

Curator notes

. Interventions/Gb: 715
. Contamination notes: "Contamination report for assembly labelled primary: Total length of scaffolds removed: 65,033,288 (2.2 %); Scaffolds removed: 567 (16.4 %); Largest scaffold removed: (6,451,661); FCS-GX contaminant species (number of scaffolds; total length of scaffolds): Pararheinheimera texasensis, g-proteobacteria (78; 2,812,616); Lacihabitans soyangensis, CFB group bacteria (32; 1,779,580); Aeromonas tecta, g-proteobacteria (30; 1,434,594); Lacunisphaera limnophila, verrucomicrobia (28; 910,335); Phnomibacter ginsenosidimutans, CFB group bacteria (27; 1,778,221); Oleiharenicola lentus, verrucomicrobia (17; 559,902); Lacibacter luteus, CFB group bacteria (16; 3,176,477); Variovorax sp. 67-131, b-proteobacteria (15; 2,498,489); Bosea vaviloviae, a-proteobacteria (14; 1,191,278); Paucibacter aquatile, b-proteobacteria (12; 1,090,731); Fuscovulum blasticum, a-proteobacteria (10; 644,917); Arcicella aurantiaca, CFB group bacteria (9; 1,464,333); Arcicella rosea, CFB group bacteria (8; 536,627); Flavobacterium hydrocarbonoxydans, CFB group bacteria (8; 285,883); Fluviicola taffensis, CFB group bacteria (8; 219,308); Lacunisphaera sp., verrucomicrobia (8; 143,246); Variovorax sp., b-proteobacteria (7; 1,586,245); Phragmitibacter flavus, verrucomicrobia (6; 186,800); Lacibacter cauensis, CFB group bacteria (5; 173,993); Runella aurantiaca, CFB group bacteria (5;

147,371); Oleiharenicola sp., verrucomicrobia (5; 130,474); Candidatus Brachybacter algidus, CFB group bacteria (4; 2,450,680); Variovorax boronicumulans, b-proteobacteria (4; 460,988); Bosea psychrotolerans, a-proteobacteria (4; 271,936); Pseudarcicella hirudinis, CFB group bacteria (4; 282,257); Flectobacillus major, CFB group bacteria (4; 270,068); Pelomonas saccharophila, b-proteobacteria (4; 267,093); Emticicia agri, CFB group bacteria (4; 154,010); Methylibium sp., b-proteobacteria (4; 134,536); Taibaiella soli, CFB group bacteria (4; 90,612); Alkanindiges illinoisensis, g-proteobacteria (3; 328,128); Taibaiella lutea, CFB group bacteria (3; 134,039); Runella rosea, CFB group bacteria (3; 206,638); Ferruginibacter sp., CFB group bacteria (3; 130,840); Terrimonas sp., CFB group bacteria (3; 183,564); Emticicia aquatica, CFB group bacteria (3; 92,793); Rheinheimera riviphila, g-proteobacteria (3; 89,477); Roseateles aquatilis, b-proteobacteria (3; 95,800); Flavobacterium geliluteum, CFB group bacteria (3; 98,798); Aquincola rivuli, b-proteobacteria (3; 43,756); Aquihabitans sp., actinobacteria (3; 47,513); Aminobacter carboxidus, a-proteobacteria (2; 5,192,793); Kinneretia asaccharophila, b-proteobacteria (2; 193,145); Acinetobacter bohemicus, g-proteobacteria (2; 188,985); Alkanindiges hydrocarboniclasticus, g-proteobacteria (2; 167,138); Bosea lathyri, a-proteobacteria (2; 179,243); Dyadobacter bucti, CFB group bacteria (2; 139,963); Xinfangfangia pollutisoli, a-proteobacteria (2; 93,184); Paucibacter toxinivorans, b-proteobacteria (2; 81,667); Niabella ginsengisoli, CFB group bacteria (2; 66,218); Pseudomonas alkylphenolica, g-proteobacteria (2; 73,890); Paucibacter sp., b-proteobacteria (2; 47,434); Candidatus Fluviicola riflensis, CFB group bacteria (2; 74,595); Dyadobacter jejuensis, CFB group bacteria (2; 65,424); Candidatus Didemnitutus sp., verrucomicrobia (2; 74,433); Paraflavitalea devenefica, CFB group bacteria (2; 70,910); Flavobacterium saccharophilum, CFB group bacteria (2; 70,356); Sediminibacterium goheungense, CFB group bacteria (2; 69,697); Variovorax guangxiensis, b-proteobacteria (2; 34,276); Stagnimonas aquatica, g-proteobacteria (2; 28,654); Rhabdothermincola salaria, actinobacteria (2; 37,829); Flavobacterium piscis, CFB group bacteria (2; 29,000); Sphingobacterium detergens, CFB group bacteria (1; 6,451,661); Devosia ginsengisoli, a-proteobacteria (1; 4,573,212); Microbacterium maritypicum, high GC Gram+ (1; 4,350,392); Sphingopyxis witflariensis, a-proteobacteria (1; 4,002,761); Mesorhizobium zhangyense, a-proteobacteria (1; 3,095,000); Flavobacterium terrigena, CFB group bacteria (1; 3,040,654); Marortus luteolus, g-proteobacteria (1; 480,538); Acinetobacter populi, g-proteobacteria (1; 376,571); Agriterribacter humi, CFB group bacteria (1; 165,394); Brevundimonas alba, a-proteobacteria (1; 126,807); Deminuibacter soli, CFB group bacteria (1; 92,967); Flavipsychrobacter stenotrophus, CFB group bacteria (1; 92,251); Chryseobacterium aahli, CFB group bacteria (1; 89,992); Microbacterium chocolatum, high GC Gram+ (1; 86,297); Pseudochelatococcus contaminans, a-proteobacteria (1; 85,121); Aquirhabdus parva, g-proteobacteria (1; 85,204); Dasania marina, g-proteobacteria (1; 84,281); Kandeliimicrobium roseum, a-proteobacteria (1; 78,619); Sinobacterium norvegicum, g-proteobacteria (1; 66,454); Bosea minatitlanensis, a-proteobacteria (1; 65,594); Dyadobacter frigoris, CFB group bacteria (1; 59,852); Panacibacter ginsenosidivorans, CFB group bacteria (1; 58,937); Pseudorhodobacter sp. PARRP1, a-proteobacteria (1; 56,388); Pseudochrobactrum algeriensis, a-proteobacteria (1; 53,645); Bosea spartocytisi, a-proteobacteria (1; 53,335); Thiomonas intermedia, b-proteobacteria (1; 51,741); Rheinheimera baltica, g-proteobacteria (1; 51,208); Bradyrhizobium brasilense, a-proteobacteria (1; 49,975); Terrimonas ferruginea, CFB group bacteria (1; 47,040); Microbacterium ihumii, high GC Gram+ (1; 45,591); Niastella vici, CFB group bacteria (1; 45,121); Sediminibacterium soli, CFB group bacteria (1; 40,015); Microbacterium phage Elva, prokaryotic viruses (1; 39,288); Flavisolibacter ginsengisoli, CFB group bacteria (1; 38,756); Brevundimonas vitisensis, a-proteobacteria (1; 38,078); Methylotenera versatilis, b-proteobacteria (1; 36,516); Cypionkella psychrotolerans, a-proteobacteria (1; 36,424); Brevundimonas lenta, a-proteobacteria (1; 36,189); Pelomonas sp., b-proteobacteria (1; 35,107); Piscinibacter defluvii, b-proteobacteria (1; 35,015); Bosea caraganae, a-proteobacteria (1; 34,972); Leucobacter weissii, high GC Gram+ (1; 34,164); Dyadobacter fermentans, CFB group bacteria (1; 33,694); Sinobacterium caligoides, g-proteobacteria (1; 31,515); Emticicia aquatilis, CFB group bacteria (1; 30,789); Flavobacterium urocaniciphilum, CFB group bacteria (1; 30,774); Dyadobacter luticola, CFB group bacteria (1; 30,110); Acinetobacter baylyi, g-proteobacteria (1; 30,098); Hydrobacter penzbergensis, CFB group bacteria (1; 30,014); Rhabdaerophilum calidifontis, a-proteobacteria (1; 29,510); Rhabdothermincola sediminis,

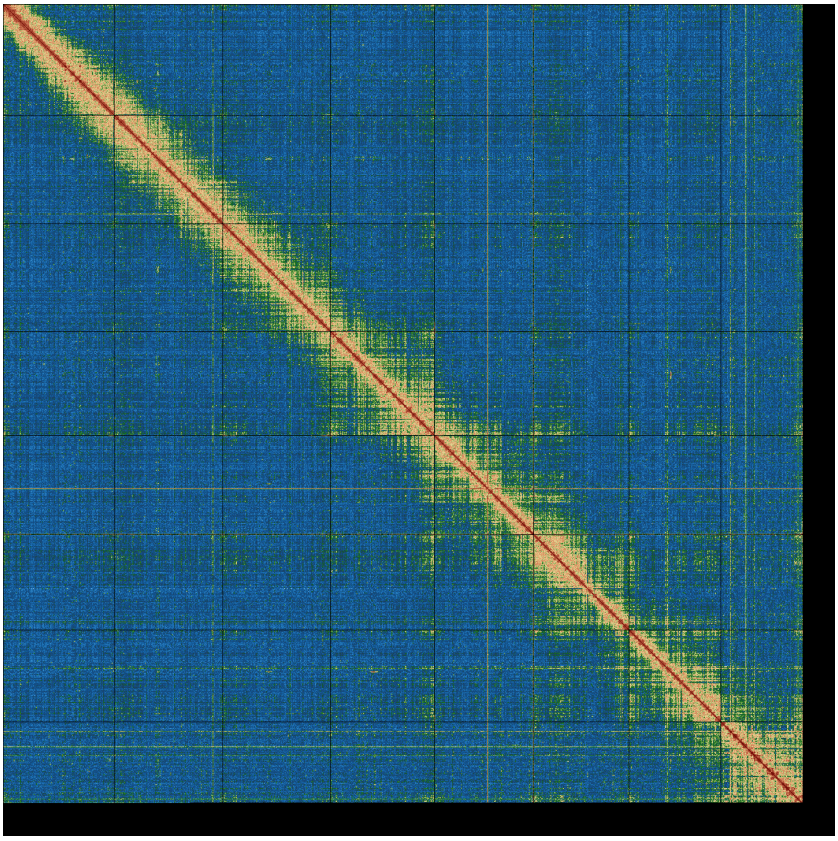
actinobacteria (1; 29,143); Acinetobacter rudis, g-proteobacteria (1; 29,010); Mucilaginibacter rivuli, CFB group bacteria (1; 28,806); Pseudogemmobacter hezensis, a-proteobacteria (1; 28,679); Rhodobacter flagellatus, a-proteobacteria (1; 27,440); Aquincola tertiaricarbonis, b-proteobacteria (1; 27,341); Dyadobacter psychrotolerans, CFB group bacteria (1; 27,016); Dyadobacter arcticus, CFB group bacteria (1; 26,917); Candidatus Vicinibacter affinis, CFB group bacteria (1; 26,691); Pedobacter planticolens, CFB group bacteria (1; 26,080); Alishewanella agri, g-proteobacteria (1; 25,768); Bacteriovorax stolpii, proteobacteria (1; 24,824); Methylotenera oryzisoli, b-proteobacteria (1; 24,207); Halobacteriovorax sp. JY17, proteobacteria (1; 23,378); Candidatus Halocyntiibacter alkanivorans, a-proteobacteria (1; 23,375); Pedobacter polaris, CFB group bacteria (1; 22,294); Pedobacter insulae, CFB group bacteria (1; 22,282); Runella zeae, CFB group bacteria (1; 21,869); Ideonella azotifigens, b-proteobacteria (1; 21,687); Halovulum dunhuangense, a-proteobacteria (1; 21,675); Cyclobacterium salsum, CFB group bacteria (1; 20,965); Chitinophaga tropicalis, CFB group bacteria (1; 19,719); Eisenibacter elegans, CFB group bacteria (1; 19,532); Rheinheimera mangrovi, g-proteobacteria (1; 19,445); Ferruginibacter lapsinanis, CFB group bacteria (1; 19,228); Taibaiella sp., CFB group bacteria (1; 17,280); Flavobacterium hydatis, CFB group bacteria (1; 17,000); Pedobacter boryungensis, CFB group bacteria (1; 16,669); Niabella agricola, CFB group bacteria (1; 16,056); Shewanella xiamenensis, g-proteobacteria (1; 15,168); Sphingomonas zeae, a-proteobacteria (1; 11,420); Paracoccus beibuensis, a-proteobacteria (1; 10,884); Foetidibacter luteolus, CFB group bacteria (1; 4,000); Variovorax gossypii, b-proteobacteria (1; 3,000); Ideonella alba, b-proteobacteria (1; 3,000); Gemmobacter aestuarii, a-proteobacteria (1; 3,000); Filimonas sp., CFB group bacteria (1; 3,000); Hydrotalea flava, CFB group bacteria (1; 3,000); Brevundimonas subvibrioides, a-proteobacteria (1; 2,000); Leadbetterella byssophila, CFB group bacteria (1; 2,000); Limnovirga soli, CFB group bacteria (1; 2,000); Aeromonas eucrenophila, g-proteobacteria (1; 2,000); Salipiger marinus, a-proteobacteria (1; 2,000); Flavisolibacter tropicus, CFB group bacteria (1; 1,000); Aquirufa nivalisilvae, CFB group bacteria (1; 1,000); Undibacterium hunanense, b-proteobacteria (1; 1,000); Arcicella sp., CFB group bacteria (1; 1,000); Sphingobacterium lumbrici, CFB group bacteria (1; 1,000); Niastella sp. SCN 39-18, CFB group bacteria (1; 1,000); Ramlibacter monticola, b-proteobacteria (1; 1,000);Mucilaginibacter arboris, CFB group bacteria (1; 1,000);Acinetobacter kyonggiensis, g-proteobacteria (1; 1,000);Acinetobacter tjernbergiae, g-proteobacteria (1; 1,000);Oceaniferula marina, verrucomicrobia (1; 1,000);Sphingobacterium soli, CFB group bacteria (1; 1,000);Pelomonas aquatica, b-proteobacteria (1; 1,000);Tabrizicola sediminis, a-proteobacteria (1; 1,000);Variovorax beijingensis, b-proteobacteria (1; 1,000);Barcodes (2; 29)"
. Other observations: ""

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	2,982,799,840	2,869,951,676
GC %	29.03	28.47
Gaps/Gbp	3,084.69	3,161.73
Total gap bp	920,100	946,800
Scaffolds	3,464	2,614
Scaffold N50	345,693,118	358,918,473
Scaffold L50	4	4
Scaffold L90	19	8
Contigs	12,665	11,688
Contig N50	600,751	603,143
Contig L50	1,369	1,332
Contig L90	5,937	5,609
QV	52.2	52.1
Kmer compl.	94.13	91.77
BUSCO sing.	92.5%	93.7%
BUSCO dupl.	3.4%	2.4%
BUSCO frag.	0.4%	0.3%
BUSCO miss.	3.8%	3.7%

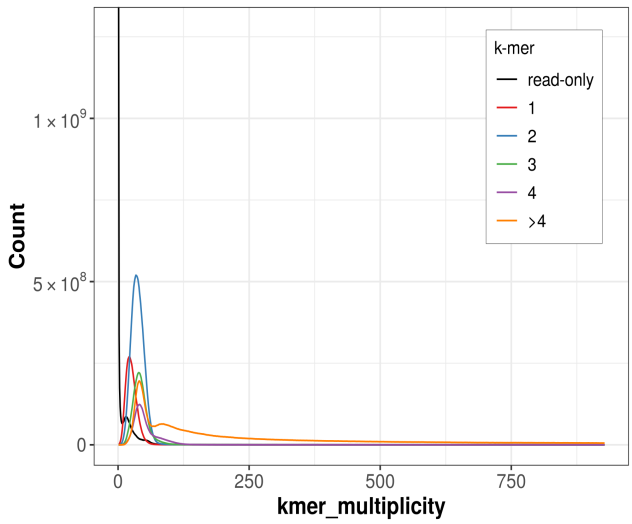
BUSCO 6.0.0 Lineage: arthropoda_odb10 (genomes:90, BUSCOs:1013)

HiC contact map of curated assembly

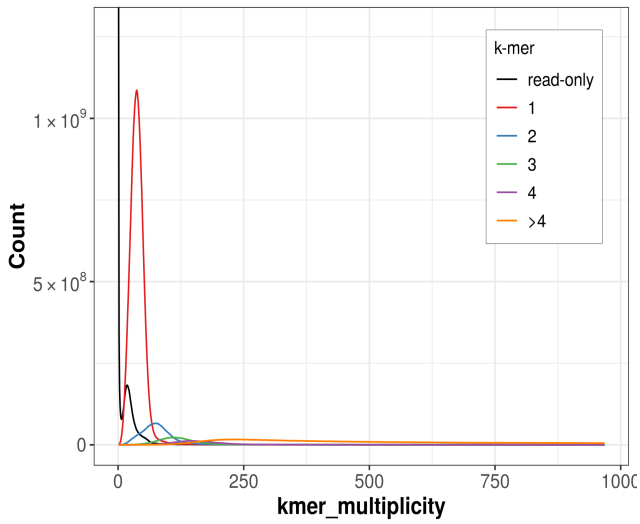


pri [LINK]

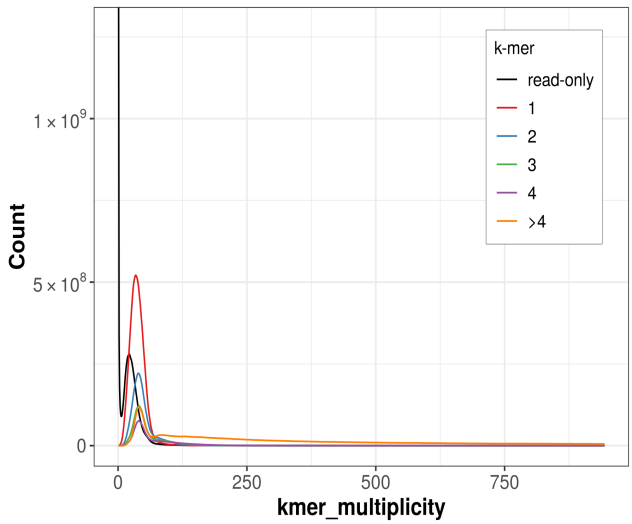
K-mer spectra of curated assembly



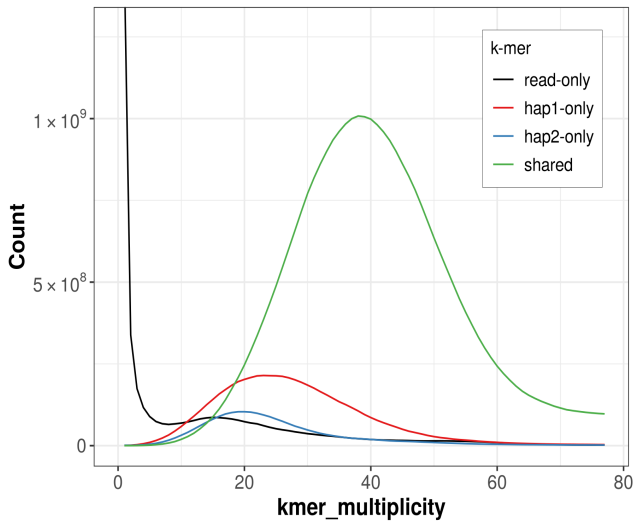
Distribution of k-mer counts per copy numbers found in asm (dipl.)



Distribution of k-mer counts per copy numbers found in **hap1** (hapl.)

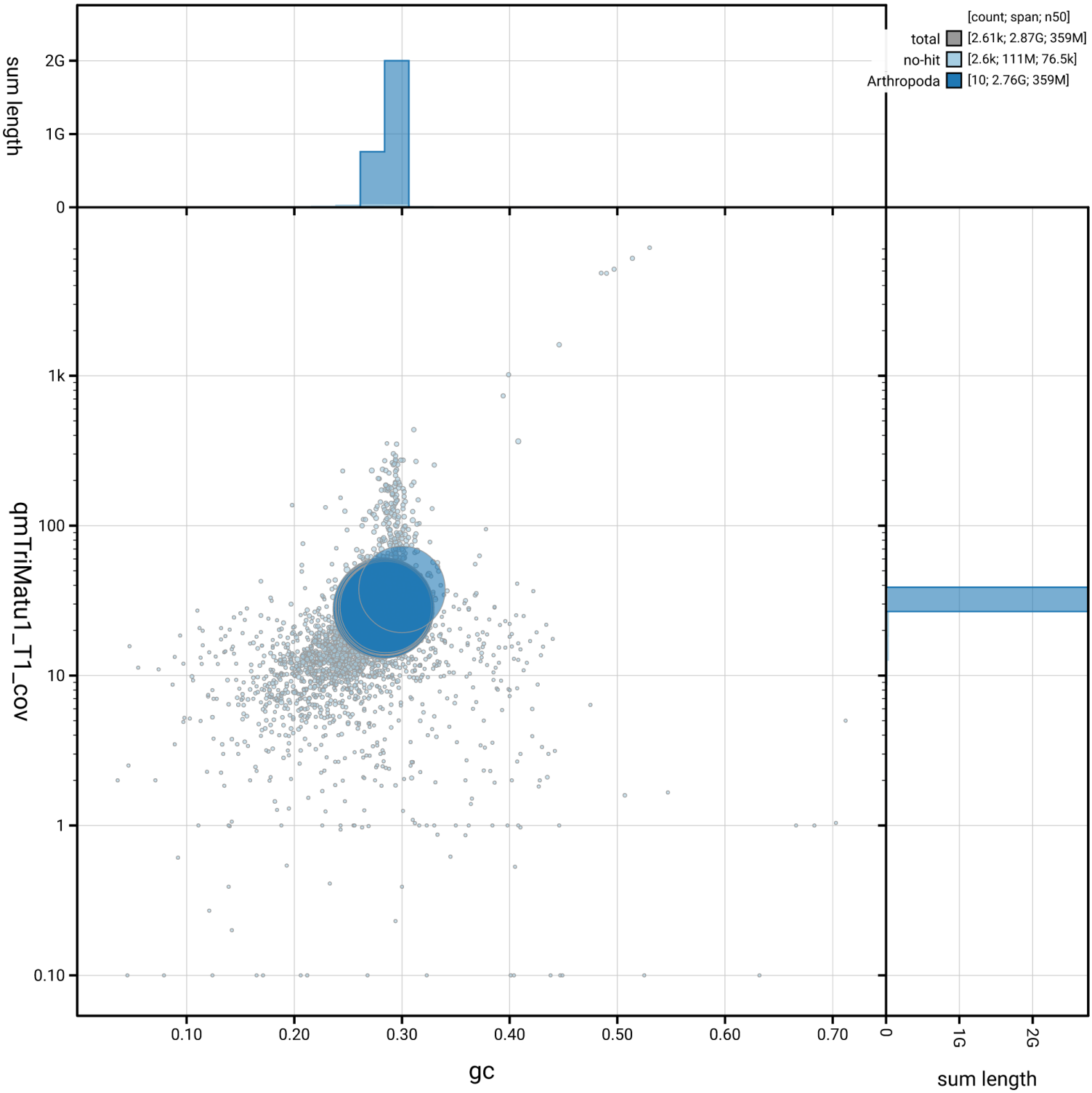


Distribution of k-mer counts per copy numbers found in **hap2** (hapl.)



Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



pri. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	PacBio HiFi	Arima v2
Coverage	37x	180x

Assembly pipeline

- **hifiasm**
    |_ *ver:* 0.25.0-r726
    |_ *key param:* --primary
- **yahs**
    |_ *ver:* 1.2.2
    |_ *key param:* NA

Curation pipeline

- **hifiasm**
    |_ *ver:* 0.25.0-r726
    |_ *key param:* --primary
- **yahs**
    |_ *ver:* 1.2.2
    |_ *key param:* NA
- **TreeVal**
    |_ *ver:* 1.2.1
    |_ *key param:* NA

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